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Agenda

* Conditional probability

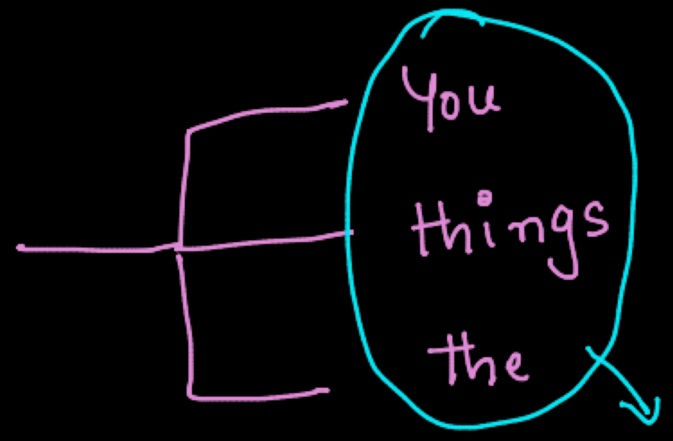
* The law of total probability

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Conditional Probability

Example : In our day to day life we will see conditional probability. So, I want to give one example.

whatsapp :



whatsapp will suggest/predict you some words

How are suggestion

What is the underlying mechanism applied either google keyboard etc

How did They specifically choose these words

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Notation

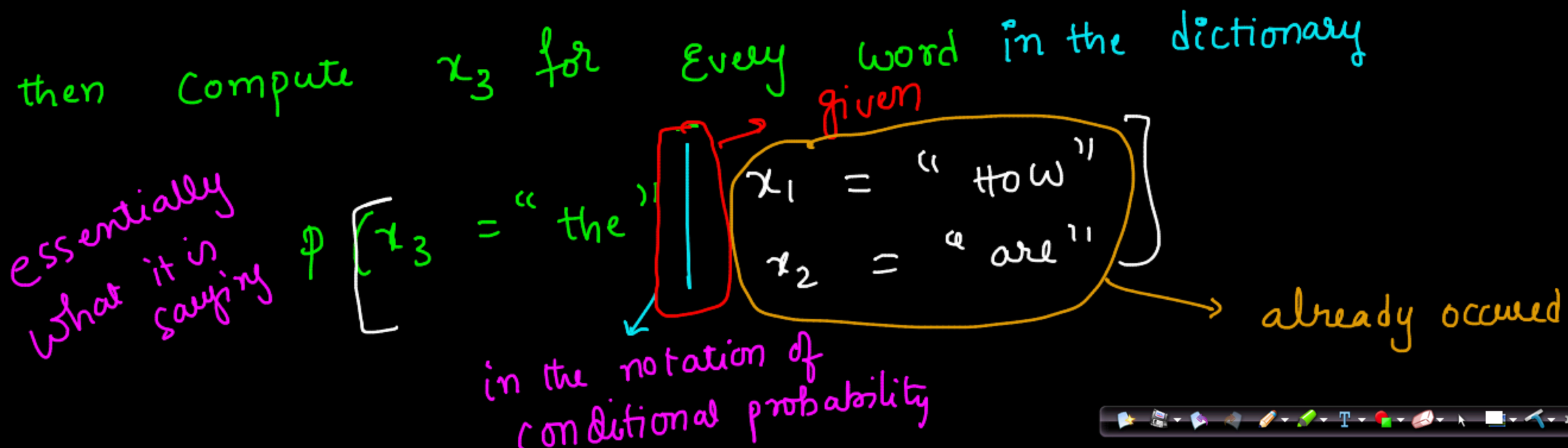
x_1 ⟶ first word

x_2 → second word

x_3 → third word

Given that x_1 = "how" and x_2 = "are" {internally the algorithm has to do following

then Compute x_3 for Every word in the dictionary



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It's computing by looking at lots of historical data.

it manages to compute the probability.

⇒ Finally it choose those words which have max probability given $\{ x_1 = \text{How}, \; x_2 = \text{are} \}$

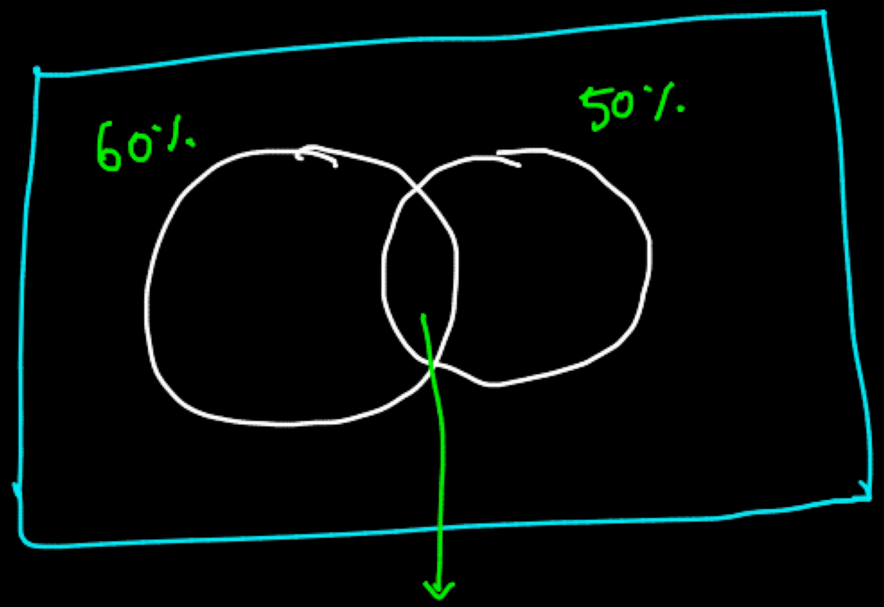
$$P(x_3 = \text{"the"} \mid x_1 = \text{"How"}, \; x_2 = \text{"are"})$$

↓ Conditional probability

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Quiz #1

It is known that 60% people use Swiggy, 50% use Zomato 20% people use both. Among those who use Swiggy, what fraction also use zomato?



Imagine there are 100 people

60 → swiggy

50 → Zomato

20 → both

→ Among 60 Swiggy users, how many use zomato also. 20 people

→ what fraction use zomato $\frac{20}{60} = \frac{1}{3}$

1/3 of swiggy users also uses zomato.

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In the language of probability will see how it will written

$P[\text{Zomato} \mid \text{Swiggy}]$

Here we are isolating the swiggy users

Who are the people collected

✓ ↓

given that person is a swiggy user

What is the probability that also going to use Zomato.

Meaning among those swiggy users, what fraction also use zomato

Layman terms

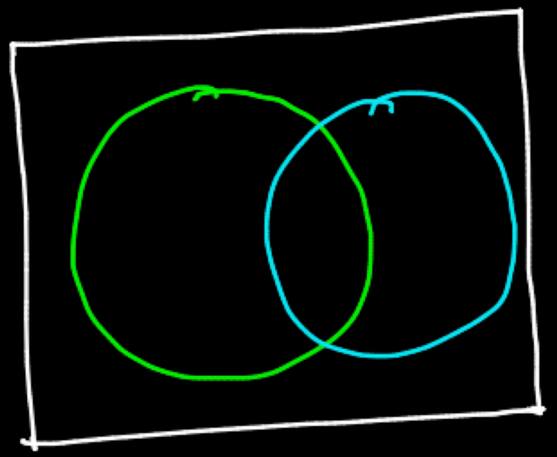
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$$P[\text{Swiggy} \mid \text{Zomato}] = \frac{20}{50}$$

Among those Zomato users, what fraction also uses Swiggy.

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Swiggy → 60%.

Zomato → 50%.

Swiggy ∩ Zomato → 20%.

① $P[\text{Zomato} \mid \text{Swiggy}] = \frac{20}{60} = \frac{P[S \cap Z]}{P(S)}$

⇓

Conditional probability

Given that a person is a Swiggy user, what is the probability that he is also uses Zomato.

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Quiz #2

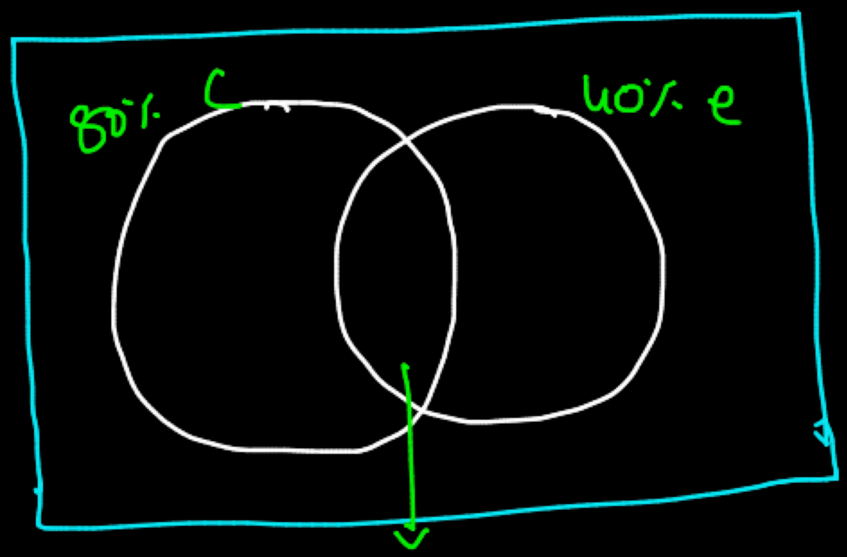
It is known that 80% people like cappuccino, 40% people like espresso, and 30% like both. Among the people who like cappuccino, what fraction of people like espresso?

0.3

30/80

30/40

0.8



$P(C) = 0.8$ $P(C \cap e) = 0.3$

$P(e) = 0.4$

$P(e \mid C) = \frac{P(C \cap e)}{P(C)}$

Quiz

$= \frac{0.30}{0.80}$

$= \frac{3}{8}$

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Quiz #3

Which of the probabilities represent the following stmt:

Among the people who like cappuccino, what fraction of people like espresso?

$P[E \mid C]$

$P[E \cup C]$

$P[C \mid E]$

$P[C \cap C]$

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Quiz #4

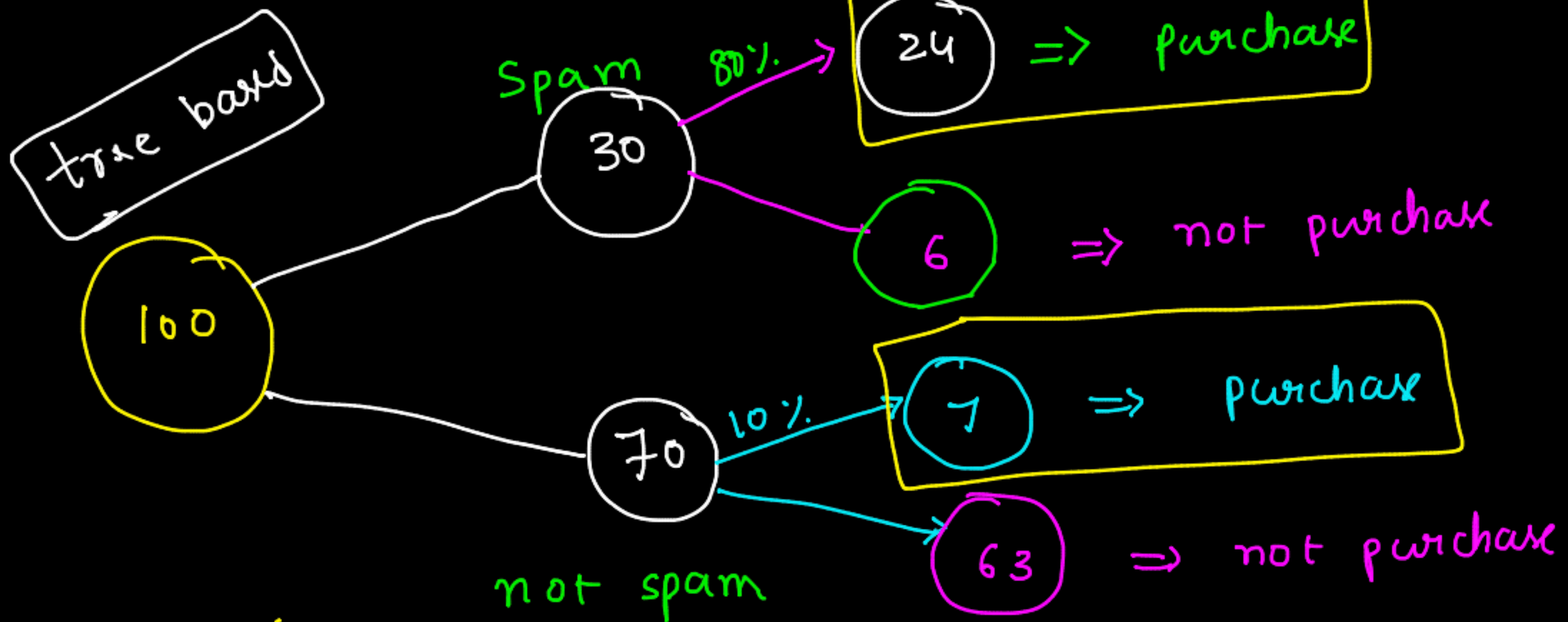
It is known that 30% of emails are spam, 70% are not spam. The word 'purchase' occurs in 80% of spam emails. It also occurs in 10% of non-spam emails. Overall, in what percentage of emails would we see the word purchase ?

45 %

10 %

✓ 31 %

71 %



30×0.8

$\frac{31}{100}$ => will have the word purchase

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Quiz #5

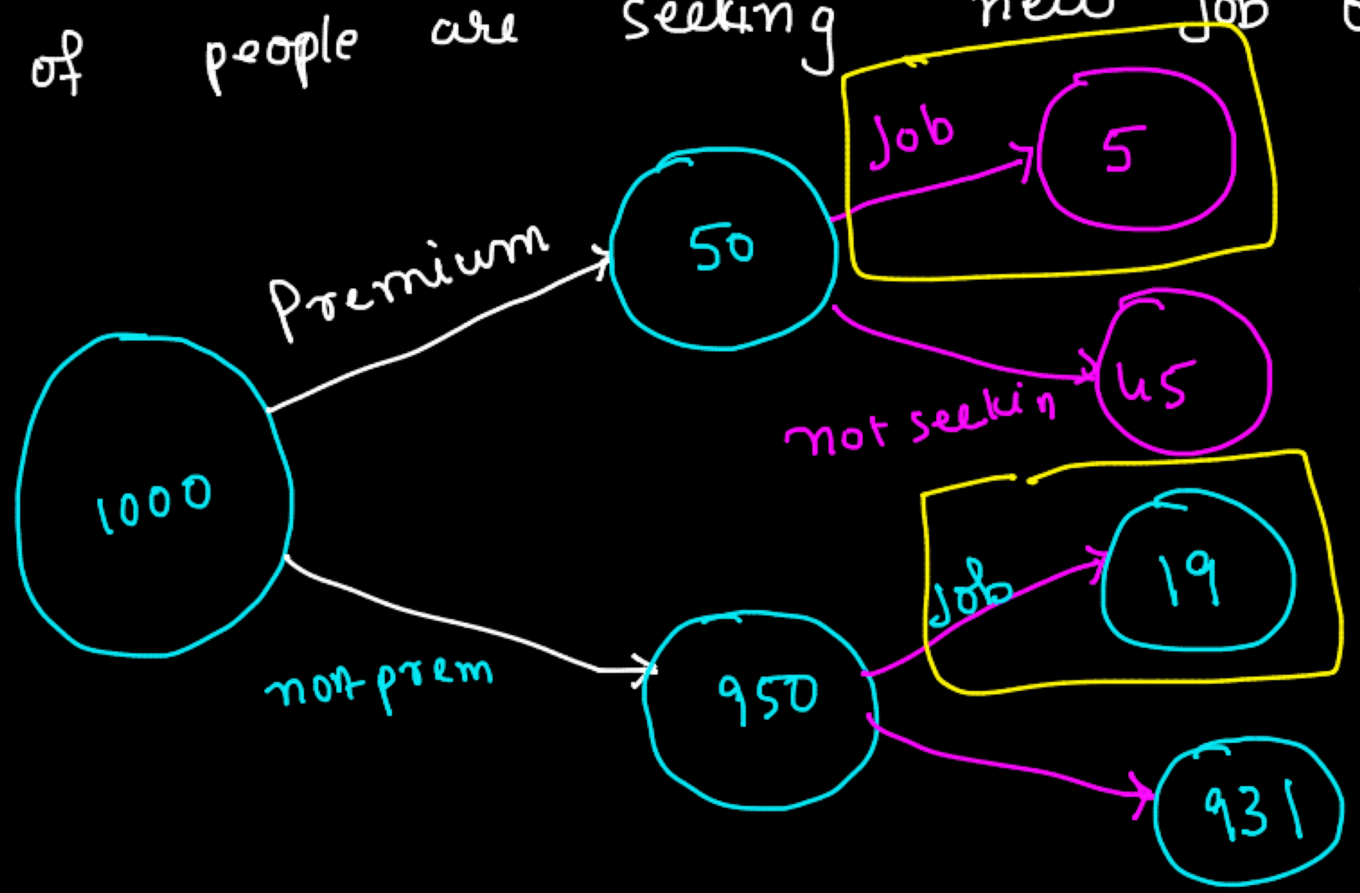
5% of all LinkedIn users are premium users. 10% of premium users are seeing new job opportunities. Only 2% of non-premium users are seeking job. Overall, what percentage of people are seeking new job opportunities?

a. 2%

b. 2.4% ✓

c. 3.7%

d. 5%



⇒ 24 are 1000 seeking job

2.4%

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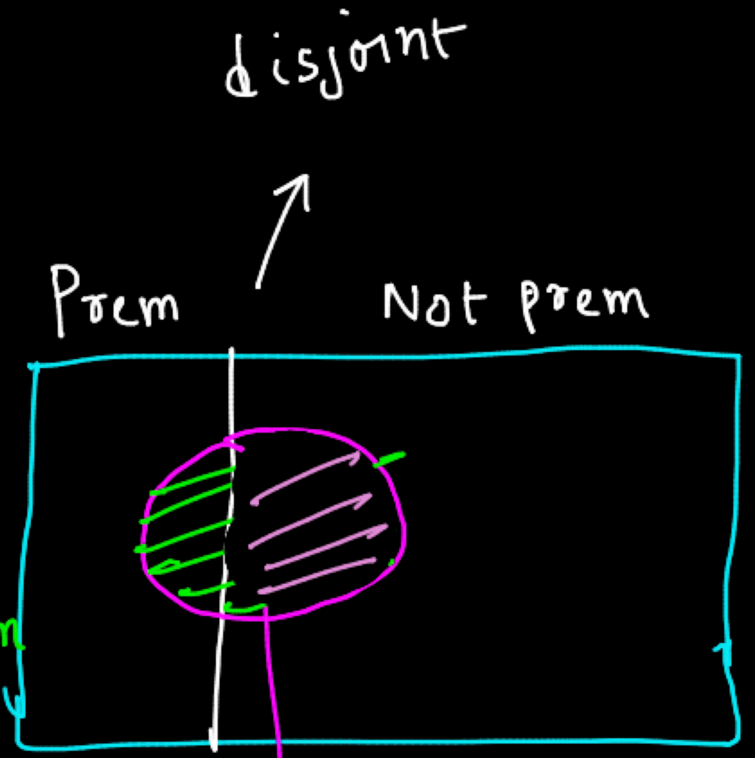
① $P[\text{Premium}] = \frac{50}{1000} = 0.05$

② $P[\text{non Premium}] = 0.95$

③ How many people are there who are seeking a job and have premium subscription.

$$P(\text{Premium} \cap \text{Job}) = \frac{5}{1000} = 0.005$$

④ $P(\text{not premium} \cap \text{Job}) = \frac{19}{1000}$



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$$P(\text{Premium} \cap \text{not Job}) = \frac{45}{1000}$$

$$P(\text{not premium} \cap \text{not Job}) = \frac{931}{1000}$$

There are conditional probability also in the tree.

$$\Rightarrow P(\text{Job seeker} \mid \text{premium}) = 10\% = 0.1$$

$$\Rightarrow P(\text{Job seekers} \mid \text{not premium}] = 2\% = 0.02$$

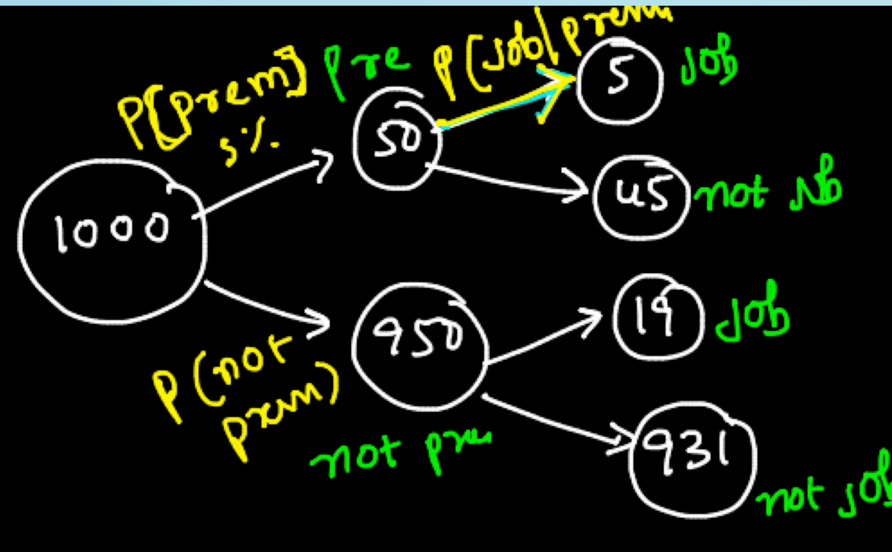
$$\Rightarrow P(\text{Job}) = \frac{24}{1000}$$

↓ probability of randomly choosen a person seeking a job

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$$P(\text{prem}) = 5\% = \frac{50}{1000}$$

$$P(\text{Job} \mid \text{premium}) = 10\% = \frac{5}{50}$$



$$P(\text{Job} \cap \text{Premium}) = P(\text{prem}) * P(\text{Job} \mid \text{prem})$$

$$= \frac{50}{1000} * \frac{5}{50} = \frac{5}{1000}$$

$$P(\text{not prem} \cap \text{Job}) = P(\text{not prem}) * P(\text{Job} \mid \text{not prem})$$

$$= \frac{950}{1000} * \frac{19}{950} = \frac{19}{1000}$$

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$$P[\text{Job}] = P[\text{job} \cap \text{Prem}] + P(\text{Job} \cap \text{not P})$$

$$= \frac{5}{1000} + \frac{19}{1000} = \frac{24}{1000}$$

law of total probability

$$P[\text{job}] = P[\text{job} \mid \text{prem}]\ P[\text{prem}] + P[\text{job} \mid \text{not prem}]$$

$$P[\text{not prem}]$$

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Quiz #6

An e-commerce website shows 2 types of ads: A & B

60% of the visitors see Type A, and 40% visitors see Type B.

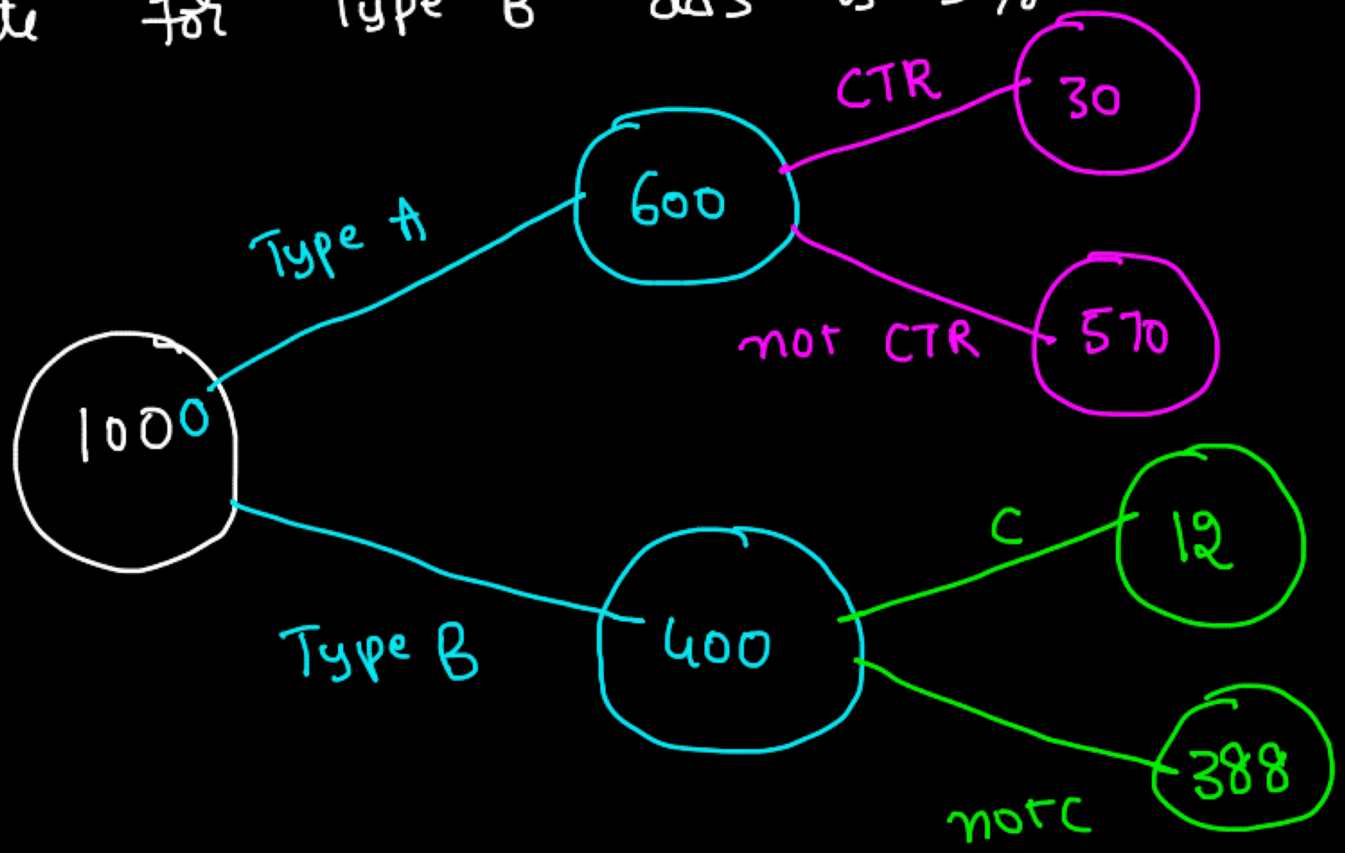
The click-through rate for Type A ads is 5%, while the click-through rate for Type B ads is 30%. What is the overall CTR?

a 3

b 4

c 4.2

d 8



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(1) $P(A) = \frac{600}{1000}$

(2) $P(B) = \frac{400}{1000}$

(3) $P(CTR \mid A) = \frac{P(C \cap A)}{P(A)}$ $\Rightarrow P(C \cap A)$

$P(C \cap A) = \frac{30}{600} * \frac{600}{1000} = \frac{30}{1000}$

(4) $P(C \cap B) = \frac{12}{1000}$

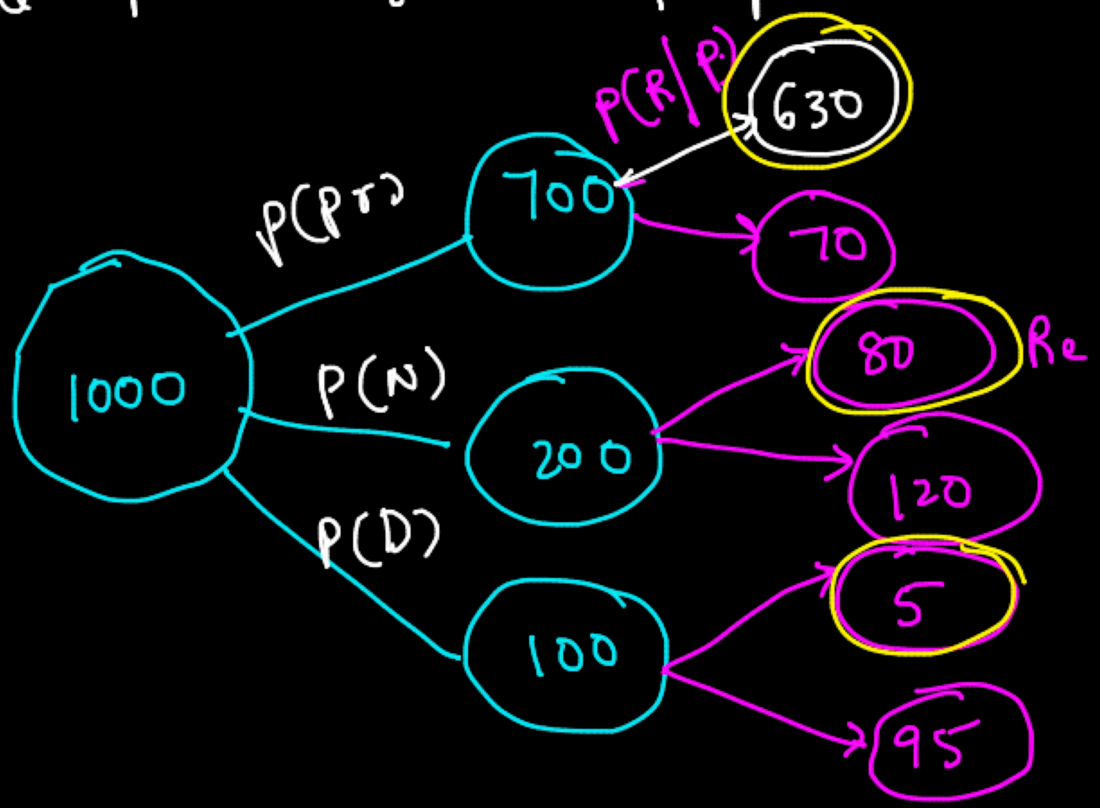
$P(B) * P(C \mid B) = \frac{400}{1000} * \frac{12}{1000}$

$P(C) = \frac{30}{1000} + \frac{12}{1000} = \frac{42}{1000}$

Quiz #7

In an NPS Survey, it is seen that 70% are promoters, 20% are neutral, 10% are detractors. 90% of promoters, 40% of neutral, and 5% of detractors recommend the product to a friend. What is the overall percentage of people who recommend the product?

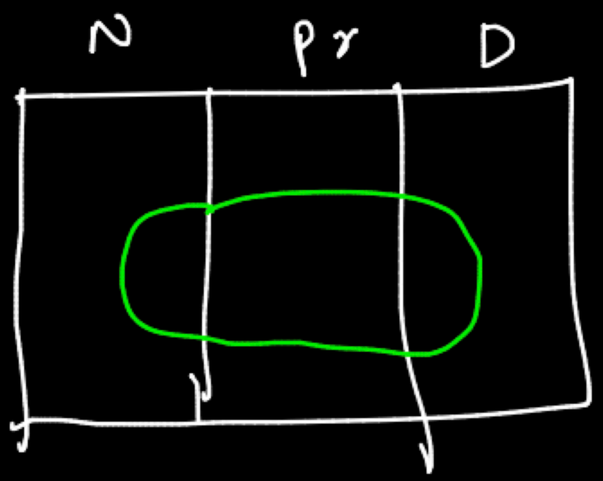
a. 50%
b 65.7%
c ✓ 71.5%
d. 82.9%



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$$P(\text{Recom}) = P(R \cap P) + P(R \cap N) + P(R \cap D)$$

$$= \frac{630 + 80 + 5}{1000} = \frac{715}{1000}$$



Mutually Exclusive and exhaustive

if you bring all these people together so whole 100% covered

exhaustive:

Pr, N, D → Pr ∪ N ∪ D has everybody

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(1) $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$ → Conditional

(2) $P(A \cap B) = P(A \mid B) * P(B)$ → Multiplication rule

(3) $P(C) = P(C \cap A)\ P(A) + P(C \cap A^c)\ P(A^c)$

→ Total probability

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